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Information Gain Constraints for Localized Dual Control

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Abstract

We show that an information constraint derived from the goal removes the probing weight that the designer of a dual controller tunes for every system and goal. The constraint acts on the Bethe free energy of a factor graph and demands information about unknown parameters and hidden states only while their uncertainty spreads the predictions beyond the goal tolerance. Message passing minimises the constrained free energy, and the Karush–Kuhn–Tucker multiplier of the constraint acts as the probing weight, which vanishes once the model suffices for the goal. We test it on a light–dark positioning task, whose good probing weights differ tenfold between a tight and a loose goal. One untuned configuration performs close to dual model predictive controllers tuned for each goal, and better than any single fixed weight.

(Claude) Add the quadcopter experiment (Section 5.1) to the abstract, the contributions and the Conclusion.

Keywords: Dual control, Bethe free energy, factor graphs, message passing, information gain

1. Introduction

Dual control refers to the problem of driving a system with unknown parameters, requiring both learning and control (Feldbaum, 1960). The optimal dual controller stems from dynamic programming, but it is intractable except in special cases (Bar-Shalom and Tse, 1974; Mesbah, 2018). Explicit dual controllers therefore add an excitation or uncertainty term to a model predictive control (MPC) cost (Mesbah, 2018; Heirung et al., 2017). However, the weight of that term sets the balance between probing and regulation, so the designer has to tune it for every system and task. MPC with integrated experiment design (MPC-X) instead imposes an information requirement that it derives from the control objective, but the requirement depends on an accuracy parameter that the designer chooses (Larsson et al., 2013).

Inference on factor graphs offers a structured alternative, because message passing computes posterior beliefs from local operations on the graph of a probabilistic model (Loeliger et al., 2007). Control as inference plans by message passing on a model whose factors encode the costs (Levine, 2018; Hoffmann and Rostalski, 2017). The Bethe free energy is a variational objective over the local beliefs of a factor graph, and belief propagation and variational message passing find its stationary points under different local constraints (Yedidia et al., 2005; Şenöz et al., 2021). Active inference adds an information gain to the goal-seeking objective through the expected free energy (EFE), but it fixes the weight of that gain by design (van de Laar et al., 2022).

We show that an information constraint on the Bethe free energy brings probing into the control posterior with a weight that the goal determines. The constraint takes its tolerance from the goal prior, which the controller needs anyway, and its Karush–Kuhn–Tucker (KKT) multiplier vanishes once the parameter-induced spread of the predictions fits within that tolerance. Since the parameter belief stays fixed within the plan, the controller is an explicit dual controller in the sense of Mesbah

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(2018). Our contribution is a dual controller whose probing weight follows from the goal. First, we formulate dual control as constrained Bethe free energy minimisation on a factor graph, with information constraints on the parameters and on hidden states whose floors come from the goal prior (Definition 3). Their KKT multipliers act as probing weights that vanish once the model suffices for the goal (Theorem 5, Lemma 6). Second, on a light–dark task, one untuned configuration costs less than every fixed weight over a tight and a loose goal.

2. Problem Statement

Consider discrete-time systems with unknown parameters,

$$x_k = f(x_{k-1}, u_k, \theta) + w_k\,, \qquad y_k = g(x_k) + v_k\,, \tag{1}$$

where $x_k \in \mathbb{R}^{n_x}$ is the state, $u_k \in \mathcal{U} = [-1, 1]^{n_u}$ the input with the actuator range scaled to $\mathcal{U}$, $y_k \in \mathbb{R}^{n_y}$ the output, and $\theta \in \mathbb{R}^{n_\theta}$ a vector of constant parameters. The process noise w_k and the measurement noise v_k are white. The controller knows the model structure, i.e., the functions f and g and the noise families, and a prior distribution over θ, but not θ itself. At time t, the controller has the data $\mathcal{D}_t = \{(u_k, y_k)\}_{k=1}^{t}$ and selects the next input u_{t+1}. Its task is to regulate the output to a target $y_* \in \mathbb{R}^{n_y}$ within a tolerance $S_* \succ 0$, so we evaluate controllers by the expected goal cost $G = \mathbb{E}\big[\sum_{k=1}^{T} \frac{1}{2}\|y_k - y_*\|^2_{S_*^{-1}}\big]$ over the initial state, the parameters and the noise. We ask how a receding-horizon controller should gather information about θ such that it minimises G.

3. Probabilistic Models

We frame dual control as a Bayesian inference problem. Let $\tau = (t+1, \ldots, t+K)$ be a finite horizon. A probabilistic state-space model over the planning horizon is:

$$p(y_\tau, x_\tau, u_\tau, \theta, x_t) = \underbrace{p(x_t)p(\theta)}_{\text{priors}} \prod_{k\in\tau} \underbrace{p(y_k \mid x_k)}_{\text{likelihood}} \underbrace{p(x_k \mid x_{k-1}, \theta, u_k)}_{\text{state transition}} \underbrace{p(u_k)}_{\text{control prior}}\,. \tag{2}$$

With $s = (y_\tau, x_\tau, u_\tau, \theta, x_t)$, the model is a product of factors,

$$p(s) \propto \prod_{a\in\mathcal{V}} f_a(s_a)\,, \tag{3}$$

where every factor f_a of the set $\mathcal{V}$ depends on a subset s_a of the variables. A Forney-style factor graph (FFG) draws every factor as a node and every variable as an edge between at most two nodes, and an equality node joins the edges of a variable with more than two factors (Loeliger et al., 2007). Unlike bipartite factor graphs, FFGs make sequential computations explicit.

Fig. 1 shows the FFG of (2) over the horizon. Since future outputs are not measured, "goal prior" factors terminate the output edges and bias the inference towards trajectories, and thereby towards inputs, that realise the goal. We encode the target of Section 2 as the Gaussian goal prior

$$p_*(y) = \mathcal{N}(y \mid y_*, S_*)\,. \tag{4}$$

The controller infers the belief of the next input u_{t+1} by message passing, with a forward sweep to the end of the horizon and a backward sweep in return. The two messages on an edge multiply to

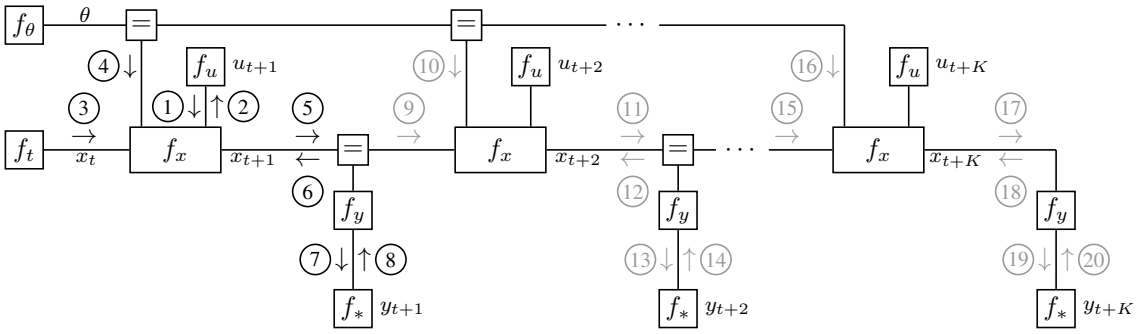


Figure 1: FFG of the planning model (2) with the goal priors (4), whose nodes are $f_t = p(x_t)$, $f_\theta = p(\theta)$, $f_u = p(u_k)$, $f_x = p(x_k \,|\, x_{k-1}, \theta, u_k)$, $f_y = p(y_k \,|\, x_k)$ and $f_* = p_*(y_k)$. Equality nodes (=) share θ and x_k between the factors that use them. On the edge u_{t+1}, the control prior sends $① = p(u_{t+1})$, and the transition node sends $② = \nu(u_{t+1})$, which it computes from the messages on its other edges.

its belief, as Fig. 1 shows for u_{t+1}. However, exact inference in (3) does not yield a dual controller. It treats the goal priors as observations, so it moves the parameter belief towards parameters under which the goal is easy to reach, an issue known as "optimism" in control as inference (Levine, 2018). Moreover, the exact posterior does not reward inputs that reveal θ, so Section 4 constrains the inference towards dual control.

4. Inference

4.1. Bethe free energy and constraints

The Bethe free energy assigns to every node of an FFG the average energy of its factor minus the entropy of its belief, and it adds the entropy of every edge belief (Yedidia et al., 2005; Şenöz et al., 2021). For the nodes $\mathcal{V}$, the edges $\mathcal{E}$ between them and a variational model $q(s)$, it reads

$$\mathcal{B}[q] = \sum_{a \in \mathcal{V}} \int q_a(s_a) \ln \frac{q_a(s_a)}{f_a(s_a)} \, \mathrm{d}s_a + \sum_{i \in \mathcal{E}} \mathbb{H}[q_i] \,, \tag{5}$$

where the beliefs q_a and q_i are the distributions of the node a and the edge i. Normalization and marginalization constraints make the beliefs valid and consistent,

$$\int q_a(s_a) \, \mathrm{d}s_a = 1 \,, \qquad \int q_a(s_a) \, \mathrm{d}s_{a \setminus i} = q_i(s_i) \,, \tag{6}$$

for every node a and every edge i of a. Under these local constraints, the stationary points of $\mathcal{B}$ are the fixed points of belief propagation, and additional constraints yield other message passing algorithms (Yedidia et al., 2005; Şenöz et al., 2021). We use two of them and add an information constraint.

Definition 1 (Factorization constraints) *The belief q_a equals the factor f_a times the edge beliefs q_i on its input edges,*

$$q_a(s_a) = f_a(s_a) \prod_{i \in \mathrm{in}(a)} q_i(s_i) \,, \tag{7}$$

where $\mathrm{in}(a)$ *is the set of input edges of node* a*.*

This structured mean-field factorization approximates the posterior but keeps every factor intact (Şenöz et al., 2021).

Definition 2 (Form constraints) *The belief* q_b *is pinned to the form of the corresponding factor* f_b*,*

$$q_b(s_b) = f_b(s_b) \quad \textit{for some } b \in \mathcal{V}\,. \tag{8}$$

For the state-space model, we pin the beliefs of the parameters and the current state to the posteriors $q(\theta) = p(\theta \mid \mathcal{D}_t)$ and $q(x_t) = p(x_t \mid \mathcal{D}_t)$, which the controller computes outside the planning model, so that the goal priors cannot cause optimism. We also pin the beliefs of the transition and measurement nodes to their factors, so that inference uses these factors as given instead of approximating them.

The information constraint bounds the information gain (Cover and Thomas, 1999) between the variables a_1 and a_2 given a third variable a_3 under a belief q_a,

$$\mathbb{I}_a[a_1 : a_2 \mid a_3] = \int q_a(a_1, a_2 \mid a_3) \ln \frac{q_a(a_1, a_2 \mid a_3)}{q(a_1 \mid a_3) q(a_2)} \mathrm{d}(a_1, a_2)\,. \tag{9}$$

For the state-space model, we constrain the information $\mathbb{I}_{t+1}[y_{t+1} : x_{t+1}, \theta \mid u_{t+1}]$ that the next output carries about the unknown state and parameters, given the next input. It follows from the joint belief of the measurement and transition nodes,

$$q(y_{t+1}, x_{t+1}, \theta \mid u_{t+1}) = q(y_{t+1} \mid x_{t+1}) \int q(x_{t+1} \mid x_t, \theta, u_{t+1})\, q(\theta)\, q(x_t)\, \mathrm{d}x_t\,. \tag{10}$$

Definition 3 (Information gain constraints) *The information gain between two variables in belief* q_a *given a third, has to exceed a lower bound* β*:*

$$\mathbb{I}_a[a_1 : a_2 \mid a_3] \geq \beta\,. \tag{11}$$

The constraint biases the control posterior $q(u_{t+1})$ towards informative inputs, and its lower bound β, the information floor, sets the strength of this bias. A low floor has no effect, whereas a high floor overrides the goal, so we derive the floor from the goal instead of tuning it. Since the goal tolerates the spread S_* around the target, no information is needed while the predictions stay within this tolerance. Let $\mu(u)$ and $\Sigma(u)$ be the mean and covariance of the predictive distribution $q(y_{t+1} \mid u_{t+1} = u)$. The covariance matters only at inputs that reach the goal, so we weight every input by its prior probability and by the goal likelihood of its mean prediction,

$$\tilde{p}_t(u) \propto p(u)\, \mathcal{N}\big(\mu(u) \mid y_*, S_*\big)\,, \qquad \bar{\Sigma}_{t+1} = \mathbb{E}_{\tilde{p}_t}\big[\Sigma(u)\big]\,. \tag{12}$$

Information is thus needed when the covariance $\bar{\Sigma}_{t+1}$ of the goal-reaching inputs exceeds the tolerance. Since no input removes the noise on the output, whose covariance R is known or estimated and enters $\bar{\Sigma}_{t+1}$, we add R to the tolerance. The floor is then the excess entropy

$$\beta_t = \mathbb{H}\big[\mathcal{N}(0, \bar{\Sigma}_{t+1})\big] - \mathbb{H}\big[\mathcal{N}(0, R + S_*)\big] = \tfrac{1}{2} \ln \big|\bar{\Sigma}_{t+1}(R + S_*)^{-1}\big|\,. \tag{13}$$

Predictions within tolerance, $\bar{\Sigma}_{t+1} \preceq R + S_*$, give $\beta_t \leq 0$ and leave the constraint inactive, whereas the further the predictions exceed the tolerance, the more information the constraint demands.

4.2. Stationary control posterior

The factorization and form constraints restrict the family of beliefs, so we substitute them into the Bethe free energy before we add multipliers for the remaining constraints (Şenöz et al., 2021). Appendix A proves Theorem 5 and Lemma 6, and the supplementary material proves Lemma 4.

Lemma 4 (Bethe collapse) *The Bethe free energy* (5) *of the model* (2) *with the goal priors* (4) *reduces, under the local constraints* (6) *and Definitions 1 and 2, to*

$$\mathcal{B}[q] = \sum_{k\in\tau} D_{KL}\big[q(u_k) \,\|\, p(u_k)\big] - \mathbb{E}_{q(y_k)}\big[\ln p_*(y_k)\big]. \tag{14}$$

In words, the transition and measurement factors cancel against their beliefs, and the input entropies of the nodes cancel the edge entropies, because every edge is the input of exactly one node. Only the divergence of the control beliefs from their priors and the expected goal log-density of the predicted outputs remain. Since (14) agrees with (5) on the constraint set, the two share their constrained stationary points. The form constraints fix $q(x_t)$ and $q(\theta)$, and the node beliefs (7) meet the marginalization at their input edges, because the transition and measurement factors are normalised conditional densities. Only the normalisation of the control beliefs and the marginalization at the outputs of the transition and measurement nodes remain. With multipliers ψ_k, λ_{y_k} and λ_{x_k} for these constraints and $\gamma \geq 0$ for the information gain constraint, the constrained Bethe Lagrangian is

$$\begin{aligned}\mathcal{L}[q] = \mathcal{B}[q] &+ \gamma\Big(\beta_t - \mathbb{I}_{t+1}\big[y_{t+1} : x_{t+1}, \theta \,|\, u_{t+1}\big]\Big) \\ &+ \sum_{k\in\tau}\Big(\psi_k\big(\int q(u_k)\,\mathrm{d}u_k - 1\big) + \int \lambda_{y_k}(y_k)\big(q(y_k) - \int p(y_k \,|\, x_k)\, q(x_k)\,\mathrm{d}x_k\big)\mathrm{d}y_k \\ &+ \int \lambda_{x_k}(x_k)\big(q(x_k) - \int p(x_k \,|\, x_{k-1}, \theta, u_k) q(x_{k-1})\, q(\theta)\, q(u_k)\,\mathrm{d}(x_{k-1}, \theta, u_k)\big)\mathrm{d}x_k\Big).\end{aligned} \tag{15}$$

This Lagrangian is to be minimized with respect to the marginal beliefs of each future action.

Theorem 5 (Control posterior) *At the stationary point of* (15)*, every input u_k has the belief*

$$q^\star(u_k) \propto p(u_k) \exp\Big(\sum_{m=k}^{t+K} \mathbb{E}_{q(y_m \,|\, u_k)}\big[\ln p_*(y_m)\big] + \gamma_k\, \mathbb{I}_k(u_k)\Big), \tag{16}$$

where $q(y_m \,|\, u_k)$ is the prediction of y_m that the marginalization constraints propagate from the input value u_k and the beliefs of the other inputs. The information gain $\mathbb{I}_{t+1}(u_{t+1})$ at the input value u_{t+1} has the weight $\gamma_{t+1} = \gamma$, and $\gamma_k = 0$ for later inputs.

In words, the approximate posterior is a product of the control prior times a pseudo-likelihood, consisting of the exponentiated cross-entropy between predictions and goals, and the potential information gained on states and parameters from predicted future observations. The cross-entropy term for a Gaussian goal prior (Eq. 4) and a prediction of y_k with mean μ_k and covariance Σ_k is

$$\mathbb{E}[\ln p_*(y_k)] = -\tfrac{1}{2}\|\mu_k - y_*\|^2_{S_*^{-1}} - \tfrac{1}{2}\mathrm{tr}\big(S_*^{-1}\Sigma_k\big) + \mathrm{const}\,. \tag{17}$$

The information gain term consists of This depends on the KKT multiplier, which is solved from the KKT condition. For an optimal belief $q^\star$ (Eq. 16),

$$\gamma \geq 0\,, \quad \mathbb{I}_{t+1}[q^\star] \geq \beta_t\,, \quad \gamma\big(\beta_t - \mathbb{I}_{t+1}[q^\star]\big) = 0\,, \tag{18}$$

where $q^\star$ depends on γ, so primal feasibility is an equation in the multiplier.

Lemma 6 (Information multiplier) *Let $\mathcal{U}$ be compact, and let $\mathbb{I}_{t+1}(u)$ be bounded on $\mathcal{U}$ and not $p(u)$-almost surely constant. With all other beliefs fixed, the expected information gain $h(\gamma) = \mathbb{E}_{q^\star(u_{t+1};\gamma)}[\mathbb{I}_{t+1}(u_{t+1})]$ is continuous and strictly increasing, and $h(\gamma) \to \mathbb{I}^{\max} = \operatorname{ess\,sup} \mathbb{I}_{t+1}$ as $\gamma \to \infty$. Hence $\gamma = 0$ if $\beta_t \leq h(0)$, γ is the unique root of $h(\gamma) = \beta_t$ if $h(0) < \beta_t < \mathbb{I}^{\max}$, and the constraint is infeasible if $\beta_t \geq \mathbb{I}^{\max}$.*

In all examples, the floor lies below $\mathbb{I}^{\max}$, if necessary by capping it there, so the multiplier is finite.

The planner solves the constraint on the first input of the plan, which it replans for every trial multiplier. Since this information gain is nondecreasing in γ, the planner brackets the smallest multiplier that meets the floor by quadrupling a trial multiplier from one, and then applies regula falsi. The floor averages over a grid of 399 inputs.

4.3. Message passing

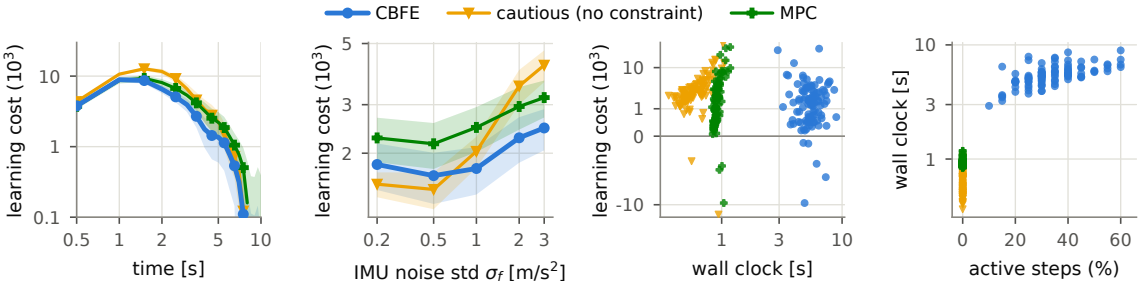
On the FFG, the control prior node sends the first factor, $p(u_k)$, along the edge of u_k, and the transition node b that consumes u_k sends the second, $\nu_{b\to u_k}(u_k)$ (Fig. 1). Message passing computes $\nu_{b\to u_k}$ backward from the goal nodes, which send p_*, because the message of every conditional node c to its input edge j is the exponentiated expectation of the log-messages on its outputs,

$$\nu_{c\to j}(s_j) = \exp\Big(\mathbb{E}_{q_c(s_c\,|\,s_j)}\Big[\sum_{l\in\mathrm{out}(c)} \ln \nu_{d_l\to l}(s_l)\Big]\Big), \tag{19}$$

where $q_c(s_c\,|\,s_j)$ is the belief (7) with s_j fixed, d_l is the node that consumes the output l, and an equality node multiplies its incoming messages. These backward messages are the exponentiated multipliers of the marginalization constraints, $\nu_{d_l\to l} = \exp(\lambda_l)$. The third factor is a local message from the information gain constraint on the edge of the next input, whose strength the multiplier sets.

which are the tracking and caution terms of dual control (Bar-Shalom and Tse, 1974), while the information gain adds probing with weight γ. For $\gamma = 0$, the mode of $q^\star$ is a cautious controller. A fixed weight on the information gain of every step gives an explicit dual MPC, and with weight one, outputs that determine the state and point-mass state beliefs, the stationary points coincide with those of EFE minimisation (van de Laar et al., 2022).

Theorem 5 defines a message passing procedure on the FFG of Fig. 1. A plan alternates a forward pass, which propagates the beliefs from the belief nodes through the conditional nodes, with a backward pass, which computes the messages (19) from the goal nodes back to the control edges and updates every control belief on its way. Since every slice repeats the transition, measurement, control and goal nodes of the first, the messages ⑨–⑭ at $t+2$ and ⑮–⑳ at $t+K$, drawn lighter in Fig. 1, are functionally equivalent, in order, to ③–⑧ at $t+1$. For a fixed multiplier, every update minimises $\mathcal{L}$ over one control belief, so the passes decrease $\mathcal{L}$ monotonically.



(*a*) Over the flight. (*b*) Over the IMU noise. (*c*) Wall clock. (*d*) Activity.

Figure 2: CBFE learns the compass fault faster than the cautious controller and avoids the bad flights of MPC. The learning cost is the goal cost minus that of an oracle that knows the wind and the fault on the same flight; ribbons show one standard error over 100 flights. (a) Learning cost per step over the flight at $\sigma_f = 2$ m/s^2, on logarithmic axes, so the curves leave the plot as the learning cost reaches zero. (b) Learning cost per step against the IMU noise. (c) Learning cost against the wall-clock time of every flight at $\sigma_f = 2$ m/s^2. (d) Wall-clock time of every flight against its share of steps with an active constraint.

5. Experiments

We test whether one untuned configuration matches dual MPCs that are tuned for each system and goal, by probing while the model is too uncertain for the goal and stopping otherwise[1]. To control for model flexibility, all controllers in a comparison share the model and the filter, and the predictive controllers share the horizon $K = 5$. We report the goal cost per step, averaged over episodes of T steps. Since all controllers share the seeds, we give paired bootstrap 95% intervals of the differences.

We call the controller with the KKT multiplier CBFE, and the cautious controller is CBFE with $\gamma = 0$. The dual MPC minimises $2J$ of (**??**) with the fixed weight $\gamma_k = \rho/2$ on every step, which comprises the tracking, caution and probing terms of explicit dual control (Mesbah, 2018; Heirung et al., 2017), so it differs from CBFE only in how it weights the information. MPC-X minimises a nominal tracking cost subject to an information requirement, whose accuracy parameter scales an application cost (Larsson et al., 2013), and we take the goal-weighted output deviation at the goal-reaching inputs as that cost. We tune ρ and the accuracy on held-out seeds by the mean RMSE, because tuning on the test seeds would favour the baselines.

5.1. Unknown parameters: quadcopter with a compass fault

A quadcopter with a faulty compass tests whether the constraint explores when caution stalls learning and exploitation misleads. An inner loop tracks the commanded thrust and attitude, so the planner commands $u_k \in [-1, 1]^3$ every $\Delta = 0.5$ s, with $K = 6$, to reach a target 5 m away and 1 m higher within $S_* = (0.03\ \mathrm{m})^2 I$ over $T = 20$ steps. Its state $x_k = (p_k, \dot{p}_k, f_k)$ holds the position,

1. Code: https://github.com/biaslab/L4DC2027-InfoConstraints

the velocity and the specific force,

$$
\begin{aligned}
f_k &= \alpha_k \begin{bmatrix} \Gamma\, n_k^{\mathrm{h}} \\ \eta\, n_k^{\mathrm{v}} \end{bmatrix} + \alpha_k \Pi_k \big(c_k - k\, \dot{p}_{k-1}\big) + w_k\,, \qquad \Gamma = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}, \\
\dot{p}_k &= \dot{p}_{k-1} + \Delta\,(f_k - g e_3)\,, \qquad p_k = p_{k-1} + \Delta\, \dot{p}_{k-1} + \tfrac{1}{2}\Delta^2 (f_k - g e_3)\,,
\end{aligned}
\tag{20}
$$

where $\alpha_k = g\,(1 + u_{k,1}/2)$ is the thrust per mass, $n_k = (\cos\varphi \sin\vartheta, -\sin\varphi, \cos\varphi\cos\vartheta)$ the thrust axis for the roll $\varphi = 0.4\,u_{k,2}$ and the pitch $\vartheta = 0.4\,u_{k,3}$, n_k^{h} and n_k^{v} its horizontal and vertical parts, and $\Pi_k = I - n_k n_k^\top$ the projection onto the rotor plane. A position fix and an IMU measure $y_k = (p_k + v_k^{\mathrm{p}}, f_k + v_k^{\mathrm{f}})$ with $v_k^{\mathrm{p}} \sim \mathcal{N}(0, 0.3^2 I)$ and $v_k^{\mathrm{f}} \sim \mathcal{N}(0, \sigma_f^2 I)$, and $w_k \sim \mathcal{N}(0, 0.2^2 I)$ collects unmodelled forces.

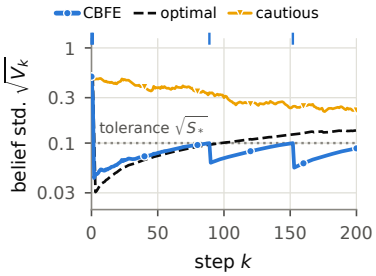
(Claude) Cite martin2010true and faessler2018differential for rotor drag, and justus1978methods for the Weibull distribution of wind speeds.

The controller learns the parameters $\theta = (a, b, \eta, k)$ and the wind drag c_k, which all enter (20) linearly. Magnetic interference rotates its heading by an unknown angle $\chi \sim \mathcal{U}[-150°, 150°]$, so that $(a, b) = (\cos\chi, \sin\chi)$ and every commanded tilt pushes in a rotated direction; this compass fault is what probing has to reveal. Rotor drag grows with the thrust and with the airspeed in the rotor plane, so k scales the drag of its own velocity and $c_k = k\,\omega_k$ the drag of the wind ω_k, whose Weibull-distributed mean speed and direction fluctuate with correlation times of 1 to 10 s. The prior on (a, b) matches the moments of the fault distribution, because only a prior that states the uncertainty lets the floor see it, and an extended Kalman filter tracks x_k and θ and models c_k as a random walk. Since the goal acts only on the position, the constraint and its floor (13) use the position output. We compare CBFE with the cautious controller and with MPC, which drops both the trace term of (17) and the constraint and thus acts on the point estimates.

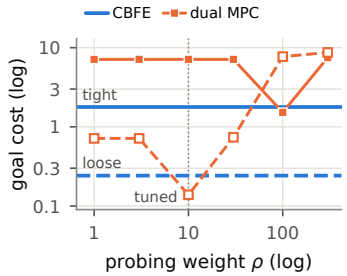
CBFE costs less than MPC at every IMU noise and less than the cautious controller from $\sigma_f = 2$ m/s^2 upward (Fig. 2(*b*)). We score the true position, since the position fix is ten times noisier than the tolerance, and we fixed the fault range and the sensor levels in a pilot on 30 separate seeds before the 100 test flights. We report the learning cost, the goal cost above that of an oracle that knows the wind and the fault. At $\sigma_f = 2$ m/s^2, CBFE attains a learning cost of 2275 per step, against 2955 for MPC and 3507 for the cautious controller, a cut of 23% and 35% with paired differences of 680 [325, 1092] and 1232 [233, 2387]. Below $\sigma_f = 1$ m/s^2, the intervals of the difference to the cautious controller include zero.

The constraint triggers when the cautious controller is over-cautious, and it turns this caution into exploration. While the rotation is unknown, every tilt spreads the predicted position, so the trace term holds the first tilt of the cautious controller at 7°, whereas the goal-reaching inputs tilt fully. The constraint is active in the first two steps of every flight and raises the first tilt of CBFE to 29°, so that the first measurements reveal the rotation, and it acts again on 26% of the later steps. CBFE therefore closes in on the target faster than the cautious controller during the first 5 s (Fig. 2(*a*)).

Pure exploitation is not cautious enough and produces occasional bad flights. MPC tilts by 31° from the first step and keeps acting on its point estimate while noisy measurements correct it, so its median cost lies only 6% above that of CBFE, whereas its worst 10% of flights cost 25% more. CBFE flies better than MPC in 42% of the flights with $|\chi| < 1$ rad and in 78% of those with $|\chi| > 2$ rad. The multiplier search raises the planning time from 45 to 271 ms per step, within the control period of 0.5 s, and the time of a flight grows with its share of active steps (Figs. 2(*c*) and 2(*d*)).



(*a*) Light–dark, tight goal.



(*b*) Light–dark, fixed weights.

Figure 3: CBFE probes only while its model is too uncertain for the goal, whereas every fixed weight costs four to 36 times as much as CBFE under one goal. (a) Belief spread in the light–dark episode of median cost; ticks mark active constraints. (b) Dual MPC per goal; 200 episodes each.

5.2. Hidden state: light–dark positioning

A light–dark task tests whether the goal sets the probing for a hidden state, whose information comes from where the agent measures (Platt et al., 2010). The agent regulates the state $x_k = x_{k-1} + u_k + w_k$, $w_k \sim \mathcal{N}(0, 10^{-4})$, to $x_* = 0$ from $x_0 \sim \mathcal{N}(0, 0.25)$, and it measures $y_k = x_k + v_k$ with $v_k \sim \mathcal{N}(0, r(x_k))$. The noise variance falls smoothly from 100 for $x \leq 0$ to 10^{-3} for $x \geq 1.5$, so the target lies in the dark. The predicted variance does not depend on the plan, so the cautious controller has no reason to visit the light. A grid filter computes the exact posterior, whose mean m_t and variance V_t form the belief node.

The state information of the next output is $\mathbb{I}_e^x(u) = \frac{1}{2}\ln(1 + A_t/r(m_t + u))$, with the variance $A_t = V_t + q$ of the next state and the noise at its predicted mean. The goal acts on the state, so the floor is $\beta_t^x = \frac{1}{2}\ln(A_t/S_*)$, and the agent seeks the light until it knows its state to within the goal tolerance. We run 200 episodes of $T = 200$ steps for a tight goal, $S_* = 0.01$, and a loose one, $S_* = 0.1$, and report the goal cost per step, $(x_k - x_*)^2/(2S_*)$. Besides the cautious controller and the dual MPC, we compare belief-space planning (BSP), which propagates the variance through the most likely measurements within the plan, and the optimal controller from dynamic programming on the Gaussian belief.

Under the tight goal, CBFE attains a goal cost of 1.78 per step, against 7.12 for the cautious controller and 6.05 for BSP, whose horizon is too short to value a visit. Fig. 3(*a*) shows that the constraint of CBFE activates whenever its belief spread reaches the goal tolerance, after which a visit to the light collapses the spread, whereas the cautious controller never localises. After the first 20 steps, CBFE keeps the state within two goal standard deviations on 97% of the steps, against 40% for the cautious controller. The optimal controller attains 1.36, 0.42 [0.23, 0.60] below CBFE, and it returns to the light 1.2 times per episode, against 3.2 times for CBFE. Under the loose goal, CBFE attains 0.24, against 0.72 for the cautious controller and 0.13 for the optimal one.

Fig. 3(*b*) shows that no single probing weight serves both goals. The dual MPC attains 0.14 with $\rho = 10$ under the loose goal, but this weight never visits the light under the tight goal, where its cost is four times that of CBFE. Under the tight goal, only $\rho = 100$ improves on CBFE, by 0.26 [0.12, 0.37], and it parks at the light under the loose goal, at 32 times the cost of CBFE. Tuning on

held-out seeds selects $\rho = 10$ for both goals, so one CBFE configuration costs less than every single weight over the two goals, whereas a weight tuned per goal costs 14% less under the tight goal and 43% less under the loose one.

6. Discussion

The information (11) has a closed form when the parameters enter the mean linearly with Gaussian noise, as in all examples, whereas nonlinear parameterisations require an approximation that perturbs both the constraint and its multiplier. The attainability of the floor relies on a parameter-induced covariance that is proportional to the noise covariance. Our planner also evaluates the floor on a grid of inputs, whose size grows exponentially with n_u.

Constraining only the next node forces the information to come from one measurement, which costs an excursion when only distant measurements are precise, as in Section 5.2. Constraints on later nodes would spread the probing over the plan, but they require a belief that learns within the plan, as in implicit dual control (Mesbah, 2018).

Closed-loop stability and the consistency of the parameter estimates after the constraint switches off remain open.

(Claude) Theorem S1 in the supplement shows that the constraint makes the regressors sufficiently exciting for the goal, not persistently exciting; consider a pointer to it here. Cite abbasi2011improved (Abbasi-Yadkori, Pál and Szepesvári, Improved algorithms for linear stochastic bandits, NeurIPS 2011) for the log-determinant bound of its part (c); the entry is not in references.bib.

7. Conclusion

We showed that a goal-derived information constraint turns a message passing planner on a factor graph into a dual controller without a probing weight. Variational calculus yields the posterior of the next input as the product of the control prior message and a message that weighs the expected goal log-density against the information gain. The KKT multiplier of the constraint switches probing on while the uncertainty spreads the predictions beyond the goal tolerance and off once it does not. On a light–dark task, one untuned configuration cost less than every fixed weight over a tight and a loose goal. Future work concerns constraints on later nodes with a belief that learns within the plan.

Generative AI Use

(Claude) Add the generative AI statement that the L4DC template asks for.

Appendix A. Proofs

Proof of Theorem 5 The form constraints fix $q(x_t)$ and $q(\theta)$, so we set the functional derivatives of (15) with respect to $q(y_k)$, $q(x_k)$ and $q(u_k)$ to zero, in the order of the backward pass.

Outputs. The belief $q(y_k)$ enters only the goal term of (14) and the constraint of y_k, so $-\ln p_*(y_k) + \lambda_{y_k}(y_k) = 0$, and the multiplier $\lambda_{y_k} = \ln p_*$ is the log-message of the goal node.

States. The belief $q(x_k)$ enters its own constraint, the constraint of y_k through $p(y_k \mid x_k)$ and, for $k < t + K$, the constraint of x_{k+1} through the next transition, so

$$\lambda_{x_k}(x_k) = \mathbb{E}_{p(y_k \mid x_k)}\big[\lambda_{y_k}(y_k)\big] + \mathbb{E}_{q_{k+1}(s_{k+1} \mid x_k)}\big[\lambda_{x_{k+1}}(x_{k+1})\big]\,, \tag{21}$$

where $q_{k+1}(s_{k+1} \mid x_k) = p(x_{k+1} \mid x_k, \theta, u_{k+1})\, q(\theta)\, q(u_{k+1})$ is the next transition belief at a fixed x_k, without the second term for $k = t + K$. With $\nu_{d_l \to l} = \exp(\lambda_l)$, the two terms are the log-messages (19) of the measurement and next transition nodes, which the equality node of x_k multiplies, so (21) is the backward pass.

Inputs. The belief $q(u_k)$ enters its divergence from the prior, its normalization, the constraint of x_k through the transition node b that consumes it and, for $k = t+1$, the information gain. The latter is linear, $\mathbb{I}_{t+1}[q] = \mathbb{E}_{q(u_{t+1})}[\mathbb{I}_{t+1}(u_{t+1})]$, because the joint belief (10) holds only factors and fixed beliefs. Hence $\ln q(u_k) = \ln p(u_k) + \mathbb{E}_{q_b(s_b \mid u_k)}[\lambda_{x_k}(x_k)] + \gamma_k\, \mathbb{I}_k(u_k) - 1 - \psi_k$, with the belief $q_b(s_b \mid u_k) = p(x_k \mid x_{k-1}, \theta, u_k)\, q(x_{k-1})\, q(\theta)$ of b at a fixed u_k.

Unrolling. Substituting (21) repeatedly, from x_k to x_{t+K}, chains the transition and measurement beliefs into $q(y_m \mid u_k)$, so that $\mathbb{E}_{q_b(s_b \mid u_k)}[\lambda_{x_k}(x_k)] = \sum_{m=k}^{t+K} \mathbb{E}_{q(y_m \mid u_k)}[\ln p_*(y_m)]$. Normalizing by ψ_k gives the Gibbs form (16) of control as inference (Levine, 2018; Wainwright et al., 2008). ■

Proof of Lemma 6 *Exponential family.* With all other beliefs fixed, Theorem 5 gives $q^\star(u; \gamma) = p(u)\, \nu_0(u)\, e^{\gamma \mathbb{I}_{t+1}(u)} / Z(\gamma)$ for the next input u, where the message $\nu_0 > 0$ collects the expected goal log-densities and is free of γ, and $Z(\gamma)$ normalises. This is an exponential family with natural parameter γ, sufficient statistic $\mathbb{I}_{t+1}$ and log-normaliser $A(\gamma) = \ln Z(\gamma)$ (Wainwright et al., 2008).

Monotonicity. The integrand of Z is bounded on the compact set $\mathcal{U}$, because $\mathbb{I}_{t+1}$ is bounded and $\nu_0 \leq (\max p_*)^K$. Hence Z is finite and positive, and differentiation under the integral gives $h = A'$ and $h' = A'' = \mathbb{V}_{q^\star}[\mathbb{I}_{t+1}]$. This variance is positive, because $\mathbb{I}_{t+1}$ is not $p(u)$-almost surely constant and $q^\star$ shares its null sets with $p(u)$, so h is continuous and strictly increasing.

Laplace principle. For $\epsilon > 0$, the factor $e^{\gamma \mathbb{I}_{t+1}}$ weighs the inputs with $\mathbb{I}_{t+1} > \mathbb{I}^{\max} - \epsilon$, which have positive prior mass, at least $e^{\gamma\epsilon}$ times more than those with $\mathbb{I}_{t+1} \leq \mathbb{I}^{\max} - 2\epsilon$. These lose their mass as $\gamma \to \infty$, so $h(\gamma) \to \mathbb{I}^{\max}$, and $h(\gamma) < \mathbb{I}^{\max}$, as $\mathbb{I}_{t+1} < \mathbb{I}^{\max}$ on a set of positive mass.

Cases. Complementary slackness (18) admits $\gamma > 0$ only if $h(\gamma) = \beta_t$. If $\beta_t \leq h(0)$, then $\gamma = 0$ is feasible, and every $\gamma > 0$ gives $h(\gamma) > \beta_t$. If $h(0) < \beta_t < \mathbb{I}^{\max}$, then $\gamma = 0$ is infeasible, and $h(\gamma) = \beta_t$ has a unique root by the intermediate value theorem and the monotonicity of h. If $\beta_t \geq \mathbb{I}^{\max}$, no multiplier is feasible, because $h(\gamma) < \beta_t$. ■

Acknowledgments

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Supplementary Material

S1. Bethe Collapse

Table S1: Contribution of every type of node in Fig. 1 to the Bethe free energy (5) under the constraints of Lemma 4. The node a of a transition or measurement has the input edges $\mathrm{in}(a)$, and the equality node passes the belief q of its input edge.

Node	Belief	Contribution
f_t, f_θ	$q = f$ (Definition 2)	0
$f_u = p(u_k)$	$q(u_k)$	$D_{\mathrm{KL}}[q(u_k) \,\|\, p(u_k)]$
f_x, f_y	$f_a \prod_{i\in\mathrm{in}(a)} q_i$ (Definition 1)	$-\sum_{i\in\mathrm{in}(a)} \mathbb{H}[q_i]$
$=$	$q(s)\,\delta(s-s')\,\delta(s-s'')$	$-\mathbb{H}[q]$
$f_* = p_*(y_k)$	$q(y_k)$	$-\mathbb{H}[q(y_k)] - \mathbb{E}_{q(y_k)}[\ln p_*(y_k)]$

Proof of Lemma 4 The proof evaluates the node terms of the Bethe free energy (5) for every type of node in Fig. 1, and it then cancels their entropies against the edge entropies. We orient every edge along the generative model (2), from the node that generates its variable to the node that consumes it. The belief nodes f_t and f_θ and the control priors f_u then have only outputs, the goal nodes f_* have only an input, and every edge is the input of exactly one node. Table S1 collects the contributions that we derive next.

Belief nodes. The form constraints pin $q(x_t) = f_t(x_t)$ and $q(\theta) = f_\theta(\theta)$, so the log-ratios of both nodes vanish, and they contribute zero.

Control priors. The node $f_u = p(u_k)$ has the single edge u_k, so its belief is the edge belief $q(u_k)$. Its term is therefore the divergence $D_{\mathrm{KL}}[q(u_k) \,\|\, p(u_k)]$ of the control belief from its prior.

Conditional nodes. A transition or measurement node a has the belief (7), so its factor cancels in the log-ratio, $\ln(q_a/f_a) = \sum_{i\in\mathrm{in}(a)} \ln q_i$. The factor f_a is a normalised density of the output given the inputs, so integrating q_a over the output leaves the product of the input beliefs. Hence

$$\int q_a(s_a) \ln \frac{q_a(s_a)}{f_a(s_a)} \,\mathrm{d}s_a = \sum_{i\in\mathrm{in}(a)} \int q_i(s_i) \ln q_i(s_i) \,\mathrm{d}s_i = -\sum_{i\in\mathrm{in}(a)} \mathbb{H}[q_i]\,. \tag{S1}$$

The transition of step k thus contributes the negative entropies of $q(x_{k-1})$, $q(\theta)$ and $q(u_k)$, and the measurement of step k contributes that of $q(x_k)$. The same integration shows that the marginals of q_a on its input edges equal the edge beliefs, so the node beliefs meet the marginalization constraints (6) at their inputs without multipliers, as Section 4 uses.

Equality nodes. An equality node has the factor $\delta(s-s')\,\delta(s-s'')$ on its three edges, and its belief is $q(s)\,\delta(s-s')\,\delta(s-s'')$, where q is the belief of its input edge. The deltas cancel in the log-ratio, so the node contributes $\int q(s) \ln q(s) \,\mathrm{d}s = -\mathbb{H}[q]$, the negative entropy of its input.

Goal nodes. The node $f_* = p_*(y_k)$ has the single edge y_k, so its belief is $q(y_k)$, and it contributes

$$\int q(y_k) \ln \frac{q(y_k)}{p_*(y_k)} \,\mathrm{d}y_k = -\mathbb{H}[q(y_k)] - \mathbb{E}_{q(y_k)}\big[\ln p_*(y_k)\big]\,. \tag{S2}$$

Cancellation. Every edge is the input of exactly one node, and this node is a conditional, equality or goal node, because the belief nodes and the control priors have no inputs. The negative

entropies in Table S1 therefore sum to $-\sum_{i\in\mathcal{E}}\mathbb{H}[q_i]$, which cancels the edge entropies of (5). For example, the edge u_k leaves f_u and enters the transition of step k, which contributes $-\mathbb{H}[q(u_k)]$, and the edge term $+\mathbb{H}[q(u_k)]$ cancels it. Only the divergences of the control priors and the expected goal log-densities of the goal nodes remain, which gives (14). ■

S2. State-Space Model: Integrator with Unknown Gain

The benchmark of Åström and Helmersson (1986) consists of $x_k = x_{k-1} + b\,u_k + w_k$ and $y_k = x_k$ that has to be driven to zero over $T = 50$ steps from $x_0 = 0$. The noise is zero-mean Gaussian distributed, $w_k \sim \mathcal{N}(0, \sigma^2)$, and the gain is unknown, $\theta = b \sim \mathcal{N}(b_0, 1)$. The input determines how much the next state reveals about the gain, and under $b_0 = 0$ the unknown sign of the gain turns a cautious controller off. The parameter belief is the Kalman filter $\mathcal{N}(\hat{b}_t, P_t)$, so the gain adds the variance $P_t u^2$ to the prediction, and the information of the transition node is $\frac{1}{2}\ln(1 + \ell_t(u))$ with $\ell_t(u) = P_t u^2/\sigma^2$. With the goal prior $\mathcal{N}(0, \sigma^2)$, the floor is

$$\beta_t = \tfrac{1}{2}\ln\frac{\sigma^2 + P_t\,\mathbb{E}_{\tilde{p}_t}[u^2]}{2\sigma^2}\,. \tag{S3}$$

We set $\sigma = 0.1$ and run 200 episodes for each prior mean $b_0 \in \{0, 0.5, 1, 2, 4\}$, with b drawn from its prior. The controllers and the tuning protocol are those of Section 5, but we report the root-mean-square error (RMSE) of the output, which orders the episodes as their goal cost does. Tuning on 50 held-out episodes per prior mean selects $\rho = 5$ from $\{0, 0.5, 1, 2, 5, 10, 20, 50\}$ and the accuracy 0.1 of MPC-X from $\{0.01, 0.03, 0.1, 0.3, 1\}$. The optimal dual controller follows from dynamic programming on the Bellman equation of Åström and Helmersson on a 121×121 grid without an input bound, and all other controllers keep $|u_k| \le 1$.

CBFE comes within 6% of the RMSE of the dual MPC with the best probing weight for every prior mean, without tuning. Over all prior means, CBFE attains an RMSE of 0.116, against 0.114 for the dual MPC with the tuned $\rho = 5$ and 0.116 for MPC-X. The paired difference to the dual MPC is 0.002 [0.001, 0.004], i.e., 2.1%, and the difference to MPC-X is 0.001 [−0.001, 0.003]. The optimal dual controller attains 0.104, but without an input bound and by minimising the expected sum of squared outputs. Outside $5 \le \rho \le 10$, the RMSE of the dual MPC exceeds that of CBFE by at least 7% (Fig. S2(*a*)). For $\rho \le 2$, it applies no input under $b_0 = 0$, where its RMSE is 0.436, and $\rho = 50$ over-probes and exceeds the RMSE of CBFE over all prior means by 24%.

Fig. S1 shows that the multiplier probes only while the gain is too uncertain for the goal. Its constraint is active on 2.0% of the steps, at the first step of every episode for $b_0 \le 0.5$, at the second step of 75% of the episodes for $b_0 = 1$, and never for $b_0 = 4$. The active multipliers have a median of 3.2, and twice this median lies inside the window of good probing weights. Without the constraint, the cautious controller applies no input under $b_0 = 0$, because every input only adds predictive spread while the sign of the gain is unknown.

S3. ARX Model: Two-Link Arm with Unknown Motor Polarity

The ARX model predicts the output from the current input and a buffer of past outputs,

$$y_k = A^\top \phi(u_k, y_{k-1}) + e_k\,, \qquad e_k \sim \mathcal{N}(0, W^{-1})\,, \tag{S4}$$

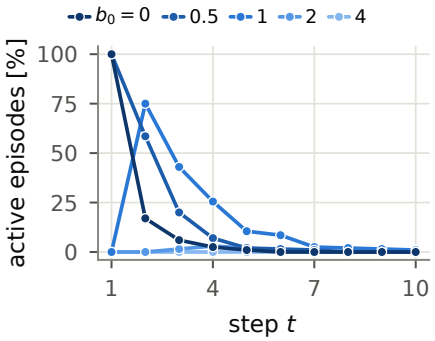


Figure S1: Share of the 200 integrator episodes per prior mean b_0 in which the information constraint of CBFE is active, per step. The constraint acts only in the first steps, and never for $b_0 = 4$.

so its state is the output buffer, and $\Theta = (A, W)$ collects the coefficients and the noise precision (Nisslbeck and Kouw, 2025). Under a matrix normal–Wishart prior, the posterior $\mathcal{MN}(A \mid M_t, \Lambda_t^{-1}, W^{-1})\, \mathcal{W}(W \mid \Omega_t, \eta_t)$ follows in closed form (Nisslbeck and Kouw, 2025; Gelman et al., 1995). Its coefficient precision accumulates the features, and the leverage of a feature vector weighs it against this precision,

$$\Lambda_k = \Lambda_{k-1} + \phi_k \phi_k^\top , \qquad \ell_t(\phi) = \phi^\top \Lambda_t^{-1} \phi \, . \tag{S5}$$

The posterior predictive is a Student-t density with covariance $(1 + \ell_t(\phi))R_t$, where R_t is the posterior mean of the noise covariance.

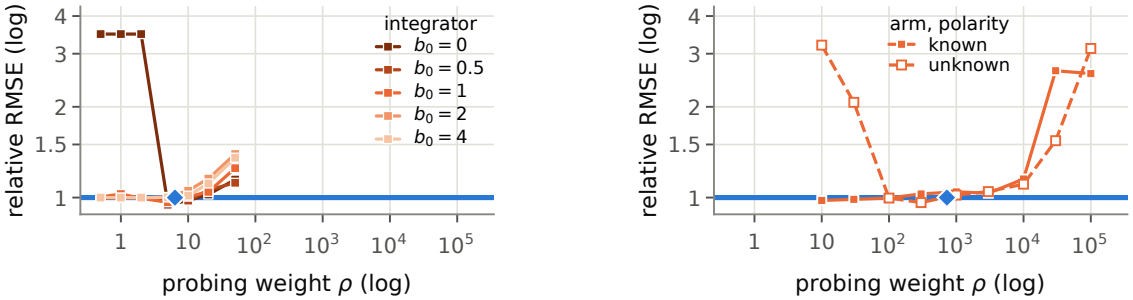
Given W, the coefficients add the covariance $\ell_t W^{-1}$ to the prediction, so their information is $\frac{n_y}{2} \ln(1+\ell_t(\phi))$, whereas the information about W does not depend on the input. We count only the coefficient information in the constraint, because the floor measures the spread that the coefficients induce, and probing cannot raise the information about W. The coefficients add the covariance $\Sigma_{t+1}^{\theta} = \ell_t R_t$ to the prediction, so the floor is $\frac{n_y}{2} \ln(1 + \bar{\ell}_t) - \frac{1}{2} \ln |I + R_t^{-1} S_*|$, with the expected leverage $\bar{\ell}_t$ of the goal-reaching inputs. As the coefficient precision Λ_t grows, this leverage falls, and the floor turns negative. The caution and probing terms of the first step are jointly minimised at the leverage

$$\ell^\circ = \max\left\{0,\, \gamma n_y / \mathrm{tr}(S_*^{-1} R_t) - 1\right\}, \tag{S6}$$

so the multiplier that a given leverage requires scales with $\mathrm{tr}(S_*^{-1} R_t)/n_y$.

The arm moves in the horizontal plane with a 0.5 kg payload and torque limits of 4 N m, and the controllers act every 0.1 s on the joint angles. The output $y_k \in \mathbb{R}^4$ holds the joint angles and velocities, and the 19 features are constant, linear and first-harmonic Fourier terms of the torques and the previous output. The prior comes from 600 excitation steps of a nominal simulator, tempered to the weight of 90 steps. The goal prior has the standard deviations 0.014 rad on the angles and 0.14 rad/s on the velocities. The floor averages over a grid of 39×39 inputs for the two torques.

Under unknown polarity, each seed reverses none, one or both motors, and the controllers replace their prior over the affected coefficients by a zero-mean prior. An episode runs through four targets in each of four rounds, $T = 480$ steps in all, and we report the RMSE of the two joint angles



(*a*) Integrator, per prior mean b_0.

(*b*) Two-link arm, per polarity.

Figure S2: RMSE of the dual MPC relative to that of CBFE against its probing weight ρ, on shared logarithmic axes. The blue line marks CBFE, which needs no tuning, and the diamond marks twice the median of its active multipliers; 200 episodes per prior mean and 10 seeds per condition.

over 10 test seeds per condition. Tuning on 6 held-out seeds per condition selects $\rho = 100$ under a known and $\rho = 300$ under an unknown polarity from $\{0, 10, 30, \ldots, 10^5\}$. Under both conditions, it selects the accuracy 10^{-4} of MPC-X from seven values between 10^{-5} and 10^{-2}.

Fig. S2 shows that the window of good probing weights lies two orders of magnitude higher on the arm than on the integrator, and that CBFE finds it without tuning on both systems. Under unknown polarity, CBFE attains an RMSE of 0.525 rad, against 0.523 rad and 0.505 rad for the dual MPC tuned with a known and with an unknown polarity. The paired differences are 0.002 [−0.022, 0.027] and 0.020 [−0.003, 0.043]. Weights up to $\rho = 30$, which include the tuned weight of the integrator, apply too little torque under unknown polarity, so the arm drifts under the process noise with an RMSE of 1.09 to 1.68 rad, as for the cautious controller. With a known polarity, CBFE attains 0.503 rad against 0.501 rad for the tuned dual MPC, whereas weights from $\rho = 3 \cdot 10^4$ upward more than double the RMSE. MPC-X attains 0.595 rad under unknown polarity, 0.071 [0.009, 0.138] above CBFE.

The multiplier supplies the probing weight that the unknown polarity requires and withdraws it once the model suffices for the goal. Its constraint is active on 10% of the steps of the first round, including the first step of every seed, and never afterwards or with a known polarity. The active multipliers have a median of 363, and twice this median again lies inside the window of good weights. By (S6), the scale $\mathrm{tr}(S_*^{-1} R_t)/n_y$ is 1 on the integrator and 46 on the arm, so the goal and the noise set the scale of the probing weight that tuning has to find. On one core, the planner takes a median of 9 ms per step on the arm, against a control period of 100 ms.

S4. Sufficient Excitation for the Goal

The information constraint does not make the inputs persistently exciting, but it makes them sufficiently exciting for the goal. Persistent excitation requires an $N \geq 1$ and an $\alpha > 0$ such that $\sum_{k=t+1}^{t+N} \phi_k \phi_k^\top \succeq \alpha I$ for every $t \geq 0$ (Åström and Wittenmark, 2013). The smallest eigenvalue $\lambda_{\min}(\Lambda_t)$ of the coefficient precision (S5) then grows without bound. The constraint asks only for the information that the goal-reaching inputs need, and Theorem S1 makes this precise for the ARX

model (S4) with constant coefficients and n_ϕ features. The integrator of Section S2 is the special case with the regressor $\phi_k = u_k$ and the precision $\Lambda_t = \sigma^2/P_t$ of the gain.

The floor of Section S3 compares the features of the goal-reaching inputs with the coefficient precision. Let $M_t = \mathbb{E}_{\tilde{p}_t}[\phi(u, y_t)\,\phi(u, y_t)^\top]$ be the second moment of the next feature vector under the goal-reaching inputs (12), so that their expected leverage is $\bar{\ell}_t = \operatorname{tr}(\Lambda_t^{-1} M_t)$. The tolerated leverage $\ell_{*,t} > 0$, defined by $(1 + \ell_{*,t})^{n_y} = |I + R_t^{-1} S_*|$, is the leverage at which the predictive covariance $(1+\ell)R_t$ has the entropy of the tolerated covariance $R_t + S_*$. It equals s/σ^2 for $R_t = \sigma^2 I$ and $S_* = sI$, so $\ell_* = 1$ on the integrator. The floor then reads

$$\beta_t = \frac{n_y}{2} \ln \frac{1 + \operatorname{tr}(\Lambda_t^{-1} M_t)}{1 + \ell_{*,t}}\,, \tag{S7}$$

and the information $\mathbb{I}_{t+1}(u) = \frac{n_y}{2} \ln\big(1 + \ell_t(\phi(u, y_t))\big)$ of the next input has to reach it.

We call the regressors $\phi_1, \dots, \phi_t$ *sufficiently exciting for the goal* with tolerance $\ell > 0$ if their precision dominates the scaled second moment of the goal-reaching features,

$$\Lambda_t \succeq \ell^{-1} M_t\,. \tag{S8}$$

Unlike persistent excitation, (S8) asks for a bounded amount of information instead of an amount that grows with t, because $M_t \preceq \Phi I$ for features bounded by Φ. It also asks for this information only in the directions in which the goal-reaching inputs have features, and a larger tolerance asks for less. Theorem S1 shows that the constraint keeps the regressors sufficiently exciting for the goal at all but logarithmically many steps.

Theorem S1 (Sufficient excitation for the goal) *Let the features be bounded along the closed loop, $\|\phi(u, y_t)\|^2 \le \Phi$ for all $u \in \mathcal{U}$ and $t < T$.*

(a) If the regressors up to step t are sufficiently exciting for the goal with tolerance $\ell_{*,t}/n_\phi$, then $\beta_t \le 0$, and the multiplier of step t vanishes. Conversely, $\beta_t \le 0$ implies that they are sufficiently exciting for the goal with tolerance $\ell_{*,t}$.

(b) If $\lambda_{\min}(\Lambda_t) \ge \Phi/\ell_{*,t}$, then $\beta_t \le 0$.

(c) If every input meets the information constraint, $\mathbb{I}_{t+1}(u_{t+1}) \ge \beta_t$ for $t = 0, \dots, T-1$, then

$$\sum_{t=0}^{T-1} \max(\beta_t, 0) \le \frac{n_y}{2} \ln \frac{|\Lambda_T|}{|\Lambda_0|} \le C_T = \frac{n_y n_\phi}{2} \ln\Big(1 + \frac{T\Phi}{n_\phi\, \lambda_{\min}(\Lambda_0)}\Big)\,. \tag{S9}$$

Hence, for every $\epsilon > 0$, the regressors are sufficiently exciting for the goal with tolerance $(1 + \ell_{*,t})\, e^{2\epsilon/n_y} - 1$ at all but at most C_T/ϵ of the steps $t < T$.

Proof of Theorem S1 *(a)* The $n_\phi \times n_\phi$ matrix $A_t = \Lambda_t^{-1/2} M_t \Lambda_t^{-1/2}$ is positive semidefinite, and its trace is $\operatorname{tr}(\Lambda_t^{-1} M_t)$. By (S7), $\beta_t \le 0$ holds if and only if this trace is at most $\ell_{*,t}$. If $\Lambda_t \succeq n_\phi M_t/\ell_{*,t}$, then $A_t \preceq (\ell_{*,t}/n_\phi) I$, so the trace of A_t is at most $\ell_{*,t}$, and $\beta_t \le 0$. Every input then meets the constraint, because the information is nonnegative, and Lemma 6 gives a vanishing multiplier. Conversely, if the trace of A_t is at most $\ell_{*,t}$, then so is its largest eigenvalue, because A_t is positive semidefinite. Hence $A_t \preceq \ell_{*,t} I$, which is equivalent to $\Lambda_t \succeq M_t/\ell_{*,t}$.

(b) Since $\Lambda_t^{-1} \preceq \lambda_{\min}(\Lambda_t)^{-1} I$ and $M_t \succeq 0$, the expected leverage satisfies $\mathrm{tr}(\Lambda_t^{-1} M_t) \leq \mathrm{tr}(M_t)/\lambda_{\min}(\Lambda_t)$. The trace $\mathrm{tr}(M_t) = \mathbb{E}_{\tilde{p}_t}[\|\phi(u, y_t)\|^2]$ is at most Φ, so $\lambda_{\min}(\Lambda_t) \geq \Phi/\ell_{*,t}$ gives $\mathrm{tr}(\Lambda_t^{-1} M_t) \leq \ell_{*,t}$, and (S7) gives $\beta_t \leq 0$.

(c) The applied input adds the regressor $\phi_{t+1} = \phi(u_{t+1}, y_t)$ to the precision, $\Lambda_{t+1} = \Lambda_t + \phi_{t+1}\phi_{t+1}^\top$, and the matrix determinant lemma gives $|\Lambda_{t+1}| = (1 + \ell_t(\phi_{t+1}))\,|\Lambda_t|$. The information of the applied input is therefore $\mathbb{I}_{t+1}(u_{t+1}) = \frac{n_y}{2} \ln(|\Lambda_{t+1}|/|\Lambda_t|)$. This information is nonnegative and at least β_t, so it is at least $\max(\beta_t, 0)$, and the sum over t telescopes to the first inequality of (S9). For the second inequality, let $\nu_1, \ldots, \nu_{n_\phi}$ be the eigenvalues of $\Lambda_0^{-1/2} \Lambda_T \Lambda_0^{-1/2}$. Their sum is $n_\phi + \sum_{k=1}^{T} \phi_k^\top \Lambda_0^{-1} \phi_k \leq n_\phi + T\Phi/\lambda_{\min}(\Lambda_0)$, and the inequality of arithmetic and geometric means gives $\ln(|\Lambda_T|/|\Lambda_0|) = \sum_i \ln \nu_i \leq n_\phi \ln(\sum_i \nu_i / n_\phi) \leq 2C_T/n_y$. Finally, every step with $\beta_t > \epsilon$ adds more than ϵ to the sum in (S9), so at most C_T/ϵ steps have $\beta_t > \epsilon$. At every other step, (S7) gives $\mathrm{tr}(\Lambda_t^{-1} M_t) \leq (1 + \ell_{*,t})\, e^{2\epsilon/n_y} - 1$, and the converse in part (a) applies with this tolerance. ■

The constraint stops asking for information once the precision reaches a finite level, so it cannot sustain persistent excitation. With a known noise covariance, the tolerated leverage ℓ_* is constant, and the precision Λ_t never decreases. Part (b) then keeps the constraint inactive at every step after $\lambda_{\min}(\Lambda_t)$ first reaches Φ/ℓ_*, whereas persistent excitation lets $\lambda_{\min}(\Lambda_t)$ grow without bound. On the integrator, $\Phi = 1$ and $\ell_* = 1$, so the constraint switches off for good once the variance P_t of the gain falls below σ^2. The consistency of the estimates after the constraint switches off, which Section 6 leaves open, therefore depends on the excitation that the regulation and the noise provide.

Part (c) bounds how often the inputs fall short of sufficient excitation for the goal. Every nat by which the floor exceeds zero costs at least one nat of information, and the information of T steps grows only logarithmically in T. The floor therefore exceeds ϵ on at most C_T/ϵ steps, a number that grows as $\epsilon^{-1} \ln T$. Fig. S1 shows the consequence on the integrator, where the constraint acts only in the first steps of an episode.

The floor (S7) derives from the goal an information requirement of the kind that MPC-X imposes. MPC-X bounds the information matrix of the coefficients from below by a multiple c of the Hessian of its application cost (Larsson et al., 2013). With the application cost of Section 5, this requirement reads $R_t^{-1} \otimes \Lambda_t \succeq c\, S_*^{-1} \otimes M_t$ at the current precision. For $R_t = \sigma^2 I$ and $S_* = sI$, the requirement with $c = n_\phi$ implies $\beta_t \leq 0$ by part (a), which in turn implies the requirement with $c = 1$. Both requirements coincide on the integrator, where $n_\phi = 1$.